THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A rectangle's length is 2 times its breadth and its perimeter is 54 cm. Its area (cm^2) is: 1

(a) 171 **(b)** 180 **(c)** 162 **(d)** 153

2. $\frac{5}{6} + \frac{3}{8} - \frac{1}{4} =$ 1

(a) $\frac{17}{24}$ **(b)** $\frac{23}{24}$ **(c)** $1\frac{1}{6}$ **(d)** $1\frac{11}{24}$

3. Two concentric circles have radii 4 cm and 9 cm. A point is 0 cm from the common centre. It lies in the ring between the circles only if its distance is more than 4 cm and less than 9 cm. How many of the distances 3, 4, 6, 9, 8, 12 cm give a point in the ring? 1

(a) 6 **(b)** 3 **(c)** 2 **(d)** 4

4. A regular polygon has an angle of rotation 40° and its perimeter is 45 cm. The side length is (cm): 1

(a) 4 **(b)** 6 **(c)** 7 **(d)** 5

5. -14 - (-9) + (22) - (-5) = 1

(a) 32 **(b)** 22 **(c)** 24 **(d)** 20

6. A square and an equilateral triangle have equal perimeters. If the square's side is 12 cm, the triangle's side (cm) is: 1

(a) 12 **(b)** 28 **(c)** 16 **(d)** 20

7. $\frac{3}{4}$ of a number minus $\frac{1}{2}$ of it is 15. The number is: 1

(a) 70 **(b)** 60 **(c)** 30 **(d)** 12

8. A rectangle's diagonal is 12 cm. The perimeter of triangle AOB where OA = OB = 6 cm and AB = 8 cm is (cm): 1

(a) 20 **(b)** 26 **(c)** 19 **(d)** 21

9. After a rotation of 135° a regular 8-gon coincides with itself. This rotation is 3 times the smallest rotation. The smallest rotation (degrees) is: 1

(a) 75 **(b)** 45 **(c)** 55 **(d)** 35

10. The sum of the integers from -5 to 9 (both included) is: 1

(a) 40 **(b)** 29 **(c)** 31 **(d)** 30

11. A rectangular field 24 m by 10 m is divided by a diagonal into two triangles. The area of one triangle (m^2) is: 1

(a) 144 **(b)** 96 **(c)** 130 **(d)** 120

12. A drum is $\frac{3}{4}$ full. 20 litres are removed and it becomes $\frac{1}{4}$ full. The capacity (litres) is: 1

(a) 50 **(b)** 40 **(c)** 39 **(d)** 41

13. **Assertion (A):** The perimeter of a rectangle 12 m long and 5 m wide is 34 m. 1
Reason (R): Perimeter of a rectangle = 2 x (length + breadth).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** $\frac{3}{4}$ and $\frac{6}{8}$ are equivalent fractions. 1
Reason (R): Multiplying the numerator and denominator by the same non-zero number gives an equivalent fraction.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** The diameter is the longest chord of a circle. 1
Reason (R): A diameter passes through the centre of the circle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. The length and breadth of a rectangle are in the ratio 3:2 and its perimeter is 60 cm. Find the sides. 2

17. $\frac{3}{4}$ of a number is 18. Find the number and its $\frac{1}{3}$. 2

18. A circle has centre O and radius 10 cm. Points are at distances 6 cm, 10 cm, 13 cm and 9 cm from O. Find how many lie inside, on and outside the circle. 2

19. A regular polygon has rotational symmetry with the smallest angle of 30° and each side 4 cm. Find the number of sides and its perimeter. 2

20. Find the sum of the integers from -6 to 10 inclusive. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A rectangular plot 40 m by 25 m is to be fenced with 3 rounds of barbed wire costing ₹12 per metre. Find the total cost. Also find the cost of ploughing it at ₹4 per m^2. 3

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22. A school has 480 students. $\frac{5}{8}$ of them are girls. $\frac{2}{5}$ of the girls play chess. How many girls play chess and how many boys are there? 3

23. Four circles of radius 3 cm are placed at the corners of a square so that adjacent circles touch. Find the side of the square joining centres and its area. 3

24. A designer wants a symmetrical logo with 5 identical petals. Find the angle between neighbouring petals, the number of lines of symmetry and the total of angles around the centre if 5 petals are placed. 3

OR

A submarine at -30 m dives 50 m, rises 40 m, then dives 25 m. Find its position after each move and the total vertical distance moved. 3

25. A square of side 12 cm and a rectangle of length 14 cm and breadth 10 cm have the same perimeter. Verify this, and find which has the larger area and by how much. 3

OR

Find the missing numerator ?: $\frac{?}{12} + \frac{1}{4} = \frac{5}{6}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Farmer Ramesh has a rectangular field 40 m long and 25 m wide. He will fence it with wire and then sow seeds. 5

(i) Find the perimeter of the field.

(ii) Find the area of the field.

(iii) Find the cost of fencing at ₹20 per metre.

(iv) 2 kg of seeds are needed for every 50 m^2. How many kg of seeds does he need?

27. Describe the construction of a rectangle with diagonal 10 cm and one side 6 cm (you may use a compass for arcs). Find the other side if it is 8 cm, check using the property that the half-diagonal is 5 cm, and find the perimeter. 5

OR

For a regular 6-gon of side 5 cm: 5

(a) find the number of lines of symmetry,

(b) the angle of rotation,

(c) the perimeter,

(d) the angle turned by a vertex when the polygon is rotated 4 times through the smallest angle.

END OF QUESTION PAPER — Total: 50 Marks

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